HLD of Messaging APP (whatsapp)

Requirements → Entities → API → Data flow

→ HLD → Deep-Dive
- ↳ HLS
- ↳ dps
- ↳ Scale

Requirements → ① functional Req ② Non functional Requirements

functional Req

① user A → user B
send / Receive msgs

② Group chats →
→ I send → Grp → all members should receive msg
→ if I am part of a group → G receives msg → I get it

③ offline → msg is lost (X)

④ ack → ✓, ✓✓, ✓✓ (blue)

⑤ support media → as an attachment

⑥ support Multiple devices → clients



② Non-functional Requirements
is Technical

Latency → 500 ms low

Guarantee delivery

→ Billion of User, High throughput →

→ msg are not stored permanently ⇒ dont store if all clients received it

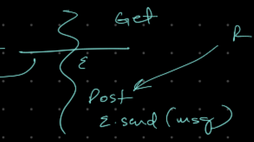
→ fault tolerant → resilient

Entities

User, client, chat, msg

flow

websockets, SSE







websocket

web socket session [ ]





→ short polling

1 sec

Latency sensitive

Not → Short Polling / Long Polling

Yes → Frequent Bi direction

No → SSE

Yes → Audio/video

No → WS

Yes → web RTC

1 web ⇒ 30 KBs

16 GB RAM ⇒ 500K

